

(Q) Suppose a grayscale image is represented as a 3×3 matrix of pixel intensities where each pixel intensity ranges from 0 (black) to 255 (white). Calculate a global threshold that can be used to separate foreground (object of interest) from the background in this image.

$$\begin{bmatrix} 120 & 135 & 145 \\ 130 & 160 & 170 \\ 125 & 140 & 150 \end{bmatrix}$$

(i) Calculate mean intensity of all the pixels in the image.

$$\text{Mean intensity} = \frac{120 + 135 + 145 + 130 + 160 + 170 + 125 + 140 + 150}{9}$$

$$= \frac{1270}{9} = 141.11 \approx 141$$

$$\text{Global threshold} = 141$$

(ii) Apply the threshold.

If pixel intensity > 141 , then pixel is 1.

If pixel intensity ≤ 141 then pixel is 0.

$$\begin{bmatrix} 120 & 135 & 145 \\ 130 & 160 & 170 \\ 125 & 140 & 150 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$